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2.2.5 Examples of unitary matrices

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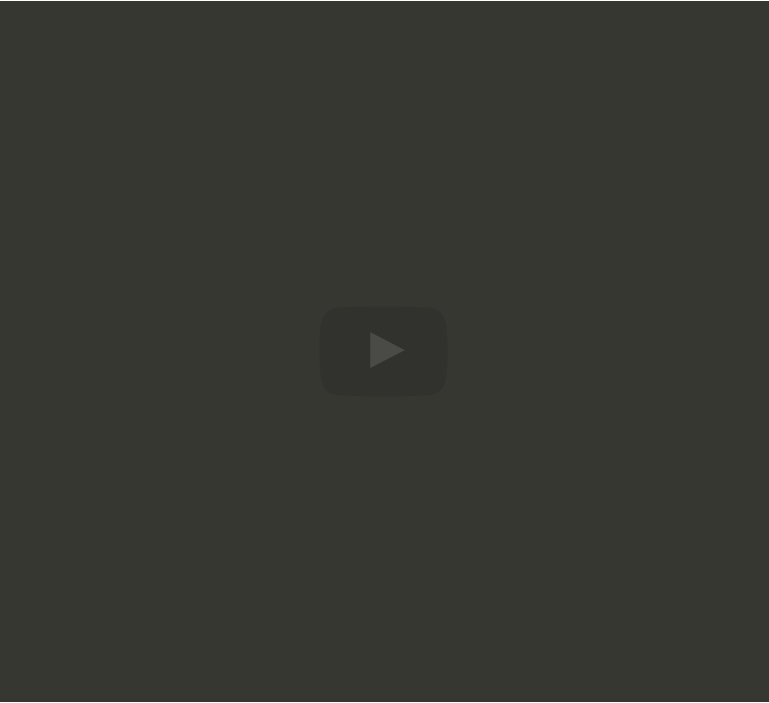
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“

is the cosine

of theta because this divided by 1 is the cosine of theta. And this right here

is the sine of theta. So this vector right here is the vector cosine

theta, sine theta. And that is what you get if you do this rotation of e_0 .

Alright? Now, what if we take e_1 through the same angle, theta? Now this is still

of length 1. And therefore this divided by that is equal to the sine of theta,

so this is sine theta. And this divided by that is the cosine of theta

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Reading assignment

1/1 point (graded)



Read Unit 2.2.5 of the notes. [\[LINK\]](#)

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Homework 2.2.5.1

1/1 point (graded)
Show that

$$\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

is a unitary matrix. (Since it is real valued, it is usually called an orthogonal matrix instead.)

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Homework 2.2.5.2

1/1 point (graded)
Prove, without relying on geometry but using what you just discovered, that $\cos(-\theta) = \cos(\theta)$ and $\sin(-\theta) = -\sin(\theta)$

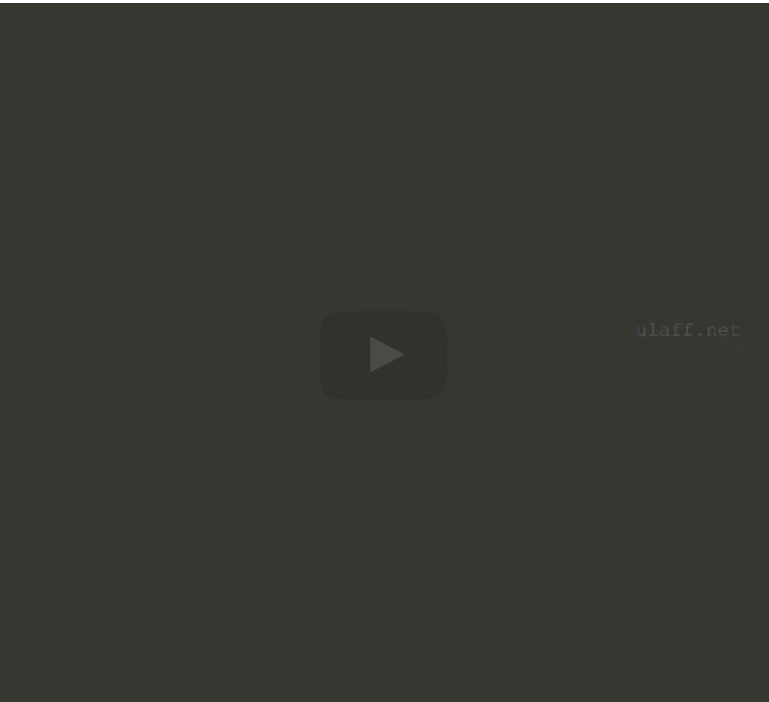
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mirror is going to have the same length as the original vector. Therefore it always preserves length. Now you can reason through that indeed reflection is actually a linear transformation. if you first scale the vector and then you reflect it you get the same thing as you would get if you first reflected and then your scaled. If you take two vectors and you add them together and then you reflect them, you get the same thing as first reflecting the two

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Homework 2.2.5.3

1/1 point (graded)
(Verbally) describe why reflecting a vector as described above is a linear transformation.

☒ done



Solution:

For complete solution, see [[LINK](#)].

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 Answers are displayed within the problem

Homework 2.2.5.4

1/1 point (graded)
Show that the matrix that represents $M : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in the above example is given by $I - 2uu^T$.

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Homework 2.2.5.5

1/1 point (graded)
(Verbally) describe why $(I - 2uu^T)^{-1} = I - 2uu^T$ if $u \in \mathbb{R}^3$ and $\|u\|_2 = 1$.

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Homework 2.2.5.6

1/1 point (graded)
Let $M : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $M(x) = (I - 2uu^T)x$, where $\|u\|_2 = 1$. Show that the matrix that represents it is unitary (or, rather, orthogonal since it is in $\mathbb{R}^{3 \times 3}$).

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Solution:

For complete solution, see [[LINK](#)].

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